Illuminate Education Case Analysis

Choosing cases of cyber conflict where stakeholders must include regulators and regulated stakeholders leads to an easily available choice of students and young people. Minors are a protected class of people worldwide. Their participation in society is typically controlled based on their age, with rights and responsibilities being afforded at different ages for things like voting, driving motor vehicles, agreeing to contracts, consuming alcohol, or joining the military. The education technology industry is highly regulated at the federal, state, and local levels due to the sensitive nature of their users, who are frequently minors, and considered a protected class. Given that today, education heavily relies on computers and software, a cyber security dispute that includes students as a stakeholder is ripe for study because of protections like these and the complexities they introduce for software producers.

Illuminate Education is a software provider for education institutions and districts, including primary and secondary schools around the country, counting millions of students as their users, in addition to teachers, administrators, and parents. Among their clients are the two largest public school districts in the country, Los Angeles and New York City, as well as districts in Colorado, Connecticut, Oklahoma, and more. These districts and schools, as well as their superior state and federal jurisdictions, employ privacy and data security privacy policies that regulate Illuminate Education as a software producer in this field, which focuses on a protected class of people. Their popular software, such as PupilPath, is used "extensively to communicate with parents and check grades for each student" (The Record, 2022). In January 2022, Illuminate Education suffered an outage to their services, which was later revealed to be related to a data breach. According to the New York Times, there was unauthorized access between December 28, 2021 and January 8, 2022. Ultimately, it was revealed that the records for at least 3 million students, across 605 districts, were accessed and possibly exfiltrated (K-12 Dive, 2023). After the breach, multiple lawsuits were filed by the parents of affected students against Illuminate Education in multiple jurisdictions, including federal court. These lawsuits included allegations of identity theft and future identity theft, harms associated with loss of privacy. However, the cases were ultimately dismissed. Contracts were also canceled with the company, and they suffered negative attention and criticism in the media and in the education technology industry. Edtech remains a lucrative space for businesses, and a target for cyberattacks, with all stakeholders working to protect their interests.

1 Context

School districts, individual schools, teachers, and students rely on software for daily school needs. Teaching, grading, scheduling, and administering all rely on software. Their use of software needs to comply with regulations to protect students' data under national and local laws. Like any other software, education software is a target for hackers, but student data also has special legal protections. Meeting those legal requirements, and a failure to do so, adds complexities that may not appear in other industries. Illuminate Education is a software producer that serves the education market in a variety of jurisdictions.

1.1 Regulatory Environment

In the US, student data can fall under multiple overlapping and changing regulations and jurisdictions that govern privacy and data security. Together, regardless of their complexity, these regulations, rules, laws, guidance, and other agreements form a regime that encompasses the regulatory requirements of a software producer. In the US, a student's data is protected federally (nationally) by the Family Educational Rights and Privacy Act (FERPA). States and other local jurisdictions protect student data through regulation as well, adding to the patchwork of regulations that software companies must adhere to. While student data protections have been enshrined in law, the enforcement of those protections is complicated. In the case of a violation of the law, individuals may not have a private right of action, this means that they cannot necessarily sue an educational institution for that violation. Instead, they may need to file a complaint with a regulatory authority, such as the Department of Education, in order to enforce the rules. Violations of these rules may also implicate other agencies, such as the Department of Justice (DOJ) or the Federal Trade Commission (FTC), if there is a commercial issue caused by the violation.

Due to the sensitive nature of this industry, there is also self-regulation in place, for example the Student Privacy Pledge, which incorporates voluntary participation to help protect student privacy. Commitment to the pledge can be legally enforceable, as the FTC and state Attorneys General "have legal authority to ensure they keep their promises" (FPF, 2020). However, as of 2022, this agreement has not led to any legal enforcement (The 74, 2022).

Student data is protected by regimes formed at the national and local levels. The regimes are established and enforced by agencies with various jurisdictions and expertise. For example, federal agencies that enforce regulations include the FTC, which is also responsible for other regulatory spaces, although the Department of Education is responsible for other education-related issues. However, local departments of education may have enforcement authority within their jurisdiction. These agencies also possess different tools for enforcement and observation based on their mandates and jurisdictions. They also have processes for creating new rules and regulations to respond to changes in the industry, threats, and violations.

Software producers in the education space are familiar with the rules and regulations that protect student data and engineer their software to comply with these requirements. They provide technical specifications to their clients to demonstrate these technical controls and practices. This dedication to the regulated arena means that they have familiarity with the types of vulnerabilities that occur within it. If a software producer wants to enter this market, having only dealt with other industries, they will need to learn the regime requirements necessary to meet the needs of their new clients. These companies may also need to comply with requirements that are unique to different jurisdictions when they sell their products and services and cooperate with clients that have varying levels of requirements.

Regulators in the education technology space include local, state, and federal government bodies, as well as non-governmental organizations that aim to protect students from a variety of harms, including privacy and data security. Their jurisdictions are not limited to cybersecurity issues, however, and can include concerns like school quality, food, transportation, safety, accessibility, and more. They may be subject to changing political climates by elected officials and economic problems that face their governments. These issues can affect their ability and their methods in observing and enforcing cybersecurity issues for their constituents and jurisdictions. One of the primary sets of rules in place is

the Federal Education Rights and Privacy Act of 1974 (FERPA). However, the mandate for federal enforcement of FERPA rules is extremely limited: Few court cases have been prosecuted, and the results of those related to privacy have been "unsuccessful," even deciding that individual students do not have the right to enforce FERPA regulations through lawsuits (Feder, 2013; Dagget 2008; Archambault, 2021). Additionally, FERPA and other student privacy regulations are seen as "inadequate" and even been "weakened" over time (Archambault, 2021; Daggett, 2008). Similarly, the Children's Online Privacy Protection Act (COPPA), meant to protect children's data as a vulnerable population, is seen as "weakly enforced and generally ineffective" (Skowronski, 2022). Furthermore, none of these laws seem to regulate a third-party software company that handles student data. State-level regulations and their enforcement mechanisms are variable. The New York City Department of Education, for example, has established New York State Education Law 2-d and Chancellor's Regulation A-820, which *inter alia* regulate the collection and use of student data by schools and third-parties, requires data privacy training for staff, and creates a "Parent's Bill of Rights" (NYC Public Schools, accessed 5/2/2025). It is feasible, nor possible, to create a full record of the regulations that constitute the regulatory environment. Instead, the aggregated rules created from statutory and case law and self-regulation are considered to comprise an abstract emergent regime that influences the regulatory environment.

1.2 Threat Environment

Threat actors are constantly changing the ways that they attempt to compromise software, exploiting new vulnerabilities, attempting new social engineering tactics, etc. Users are also implementing heterogeneous practices that can introduce new vulnerabilities. Software producers must be able to control for this variable threat surface in their engineering.

The education space is rife with breaches and creates a particularly complex problem. Since 2005, US schools have suffered nearly 2,700 breaches and 32 million records (Interview F001; Comparitech, 2023). Teachers and schools have limited budgets and are constantly trying to offer students compelling curricula, and so are frequently looking for new tools. With hundreds or thousands of end-users, trying multitudes of software tools to meet their needs, F001 contends that education may have the largest threat surface of any industry. Teachers, students, and administrators may be deploying software that meets their needs but introduces new vulnerabilities. For example, a teacher may employ a web-based tool that collects the data of its student users.

Students represent a protected class of citizen, and grade school students are also typically minors, which means they are concurrently protected. From a cybersecurity standpoint, their data is especially valuable, as it can be years before they find out that their data has been compromised, giving hackers more time to use the data for nefarious purposes (K12 Six, 2022).

Students are not the only end-users of the software; teachers, school administrators, and school district staff are all responsible for procuring and implementing the software and participate in its day-to-day use. This staff includes information technology professionals who are responsible for cybersecurity. All of these users can introduce heterogeneous practices not necessarily foreseen by the software producer or cybersecurity professionals. These heterogeneous practices may introduce new vulnerabilities that compromise the data security and privacy of all users, in particular, the protected students.

When it comes to protecting against threats to student privacy and data security, regulators' experience will vary, and their ability to guide and regulate the deployment of software will also be based on their familiarity with cybersecurity vulnerabilities in this field. They must be prepared to help prevent, respond to, and support recovery from cybersecurity issues. Based on this experience there are established regulations that aim to support cybersecurity for the student and education population.

1.3 Operational Environment

According to the New York Times, Illuminate Education claims to support more than 17 million students in 5,200 school districts in the US (2022). It supplies multiple applications for these schools and their constituents, including grade and attendance monitoring, as well as other classroom tools.

Software producers that implement software in this industry must be ready to comply with the multitude of overlapping requirements of their clients, defend against the vulnerabilities endemic to this industry, and meet their clients' needs at an affordable price to stay commercially viable. New features, changing threats, and complex regulations all add to the cost of implementing software. However,

Regulatory bodies, whether they are governmental or non-governmental, must be prepared to support the compliance efforts of their constituents, which include both the users and the software producers. This means that they provide rules and regulations that protect their constituents but don't encumber software producers such that they cannot do business. They must observe and enforce compliance with the law within the boundaries set by their elected officials' policies, judges' decisions and within their budget and capacity. In education privacy and data security, these inputs and outputs to regulatory operations can vary widely in their scope and effectiveness. For example, the NYC Department of Education may have more resources to help prevent regulatory violations compared to a small school district in Oklahoma that uses the same software. Additionally, as discussed in section 1.1., the federal government has limited operational ability to enforce protections for students; Case and statutory law do not support active enforcement methods for violations of federal regulations (Daggett, 2008).

2 Operations & Precursors

Below are the general practices and specific activities undertaken by each stakeholder leading up to the incident. The net effects of these practices and activities are driven by their outputs, which also serve as inputs to the incident that occurred.

2.1 Software Producer

Illuminate Education is a privately owned software company that provides software to school districts and individual schools. The software producer, using knowledge and expertise about their clients and this industry, would create and improve products to meet their business goals. Prior to the breach, and to this day, the company's applications are used by students, teachers, and administrators and contain protected information about students. The company itself had to secure its software, including preparing for, defending against, and preventing cyber attacks to maintain a commercially viable product and comply with regulations. Securing software is an ongoing process which can require the producer to

develop patches for their clients so they can update deployed software to protect against new and changing threats. The implementation of these patches, including the creation by the company and the dissemination to their clients, is considered a mandatory practice in software engineering. There are other evolving best practices that software producers must implement to maintain security which are supported by a massive industry that provide consulting, guidance, and additional technical tools.

Illuminate Education was founded in 2009 and developed edtech software. In 2018 and 2019 it merged with or acquired another six companies, including well-known firms like FastBridge and IO Education (Crunchbase, 2025; Sullivan, Edsurge, 2019). IOeducation is the company that originally developed PupilPath, and was itself the result of mergers and acquisitions of four other companies between 2015 and 2017 (LLR Partners, 2015). To manage this growth and meet the needs of its customers, Illuminate Education had to extend its disparate software products, such as by creating a single sign-on capability, and "create a consistent interface and user experience across those tools." This period of growth also included layoffs, estimated at about 50 people in the 600-person organization, in 2019 (Sullivan, 2019). As of 2025, however, the login area of the Illuminate Education website still provides separate links to its various applications, including IO Education, FastBridge, and others (Illuminate Education, Login, 2025). While there was no specific disclosure, PupilPath and other IO Education products were the most referenced software solutions in use by the affected customers, and were the only ones referenced later in court documents.

In providing the software to their clients, Illuminate had to comply with national and local regimes to secure its unique implementation, including preparing, defending, and preventing against cyber attacks that might compromise data, including the protected student data. However, their commercial viability and legal liability in complying with regulations varied by client need and jurisdiction. For example, in regard to commercial viability, if Illuminate offers a less expensive product than a competitor, a client may be more willing to forego cybersecurity requirements to save costs. Additionally, regarding legal liability, adherence, and compliance depending on the regime under which they were operating, the school district or school may be responsible or liable for regulatory compliance, not the software producer.

In this case, Illuminate Education provided their software PupilPath, among others, to schools and school districts across the country. This includes large districts like New York and Los Angeles which have established local regulatory regimes that govern the procurement and implementation of software within their jurisdiction. To win the business of these clients, Illuminate Education would have had to agree to comply with the rules set forth by the regime under which they were operating. Each contract would have constituted its own regime unique to the agreement with the client and the requirements of the multiple levels of jurisdictions. Illuminate Education also participated in industry self-regulation, specifically, the Student Privacy Pledge. The Student Privacy Pledge is a program administered by two software industry organizations, the Future of Privacy Forum (FPF) and the Software & Information Industry Association (SIIA) to "safeguard student privacy regarding the collection, maintenance, and use of student personal information."

Unfortunately, in this case, the software provider's operations led to unaddressed failures & weaknesses that compromised the data privacy and security of its clients. Despite their attempts at adherence/compliance, as well as their knowledge and expertise, they left vulnerabilities open for a breach. According to publicly available information, the breach was not caused by any vulnerabilities

introduced by the users. The vulnerability that was exploited has not been explicitly verified by Illuminate, but all media coverage and legal complaints point to an operational security failure exposed by the software producer. There is some evidence that the vulnerability that was exploited was specifically identified as a requirement by the NYC Department of Education: encrypting student information while the data was at rest (The Record, 2022).

2.2 Regulators

Regulatory bodies, including government agencies and self-regulatory organizations, would also be performing their duties. Depending on the context, this could include creating new rules, monitoring stakeholders, enforcing rules, and working with stakeholders towards these capacities.

The agencies charged with the development and enforcement of these regulations also have their own dynamics in achieving their goals. Local (including state and sub-state governments in the US) and national entities have different roles and responsibilities in governing the implementation and deployment of software. Among the users and software producers there are people tasked with ensuring that the software adheres to regulations, so they can be seen to be higher up in the hierarchy of user stakeholders because there are fewer of them, they have more responsibility, and they may not be as closely connected as those responsible for implementing and deploying the software.

An example of a regime that governed Illuminate Education's software development, implementation, and deployment, is with the NYC DOE. In 2020 the company "signed a strict data agreement with the [New York City] district requiring the company to safeguard student data and promptly notify district officials in the event of a data breach" (NY Times, 2022). Concurrently, Illuminate and its client the NYC DOE, signed onto the independent industry guidance program, the Student Privacy Pledge, along with other companies and school districts (NY Times, 2022; Student Privacy Pledge). This is also concurrent with any other rules established by the city and state of New York, as well as the US federal government. Ultimately, the combination of these requirements from each entity formed a unique regime that Illuminate had to comply with. The NYC DOE offers a particularly useful example because it appears publicly to have the strictest requirements on their edtech providers, but not all jurisdictions had the same requirements in place, based on publicly available documents.

An example of a self-regulatory body is the Student Privacy Pledge, which "is a public and legally enforceable statement by edtech companies to safeguard student privacy" (FPF, 2020). The pledge has hundreds of voluntary participants, and has a few regulatory impacts. First, in order to join, participants must comply with the rules of the organization, which means they may alter their processes to best practices when before they may have not. Because different jurisdictions have different requirements, this means that signatory companies may also be adhering to common, but not ubiquitous regulations. Lastly, the pledge also allows other regulatory authorities, like the FTC and state AGs to ensure that signatories comply through enforcement (FPF, 2020).

2.3 Users

In deploying this software, and others, the users (including districts, schools, teachers, and students) would have gained knowledge and expertise with the software. During the life of the software they also

would have needed to apply patches to protect against new and changing cybersecurity threats. These changes, when possible, should have been documented for users to implement properly. Additionally, when these changes fell under regulatory regimes, should be documented for adherence and compliance.

The sale, procurement, and deployment of this software would require the involved participation of various people from both the school district who procured the software, and the software producer who provided it. They would have gone through countless conversations and steps to tailor the software to meet the needs of the school district's various users. In procuring and implementing the software, clients also had to comply with national and local regimes to secure its unique implementation, including preparing, defending, and preventing against cyber attacks that might compromise their data, including the protected student data.

All these actions, which were undertaken by the stakeholders, would be documented in artifacts. Procuring the software required signed agreements, between the software producer and user. Regulations were coded in law by agencies. Deployment for day-to-day operations would be documented. Logs would be created through regular use of the software. These artifacts can be made available for both daily use, and future forensic analysis should some conflict occur.

3 Incident & Impact

Between December 28, 2021 and January 8, 2022 Illuminate Education suffered a breach that targeted the data of as many as 3 million student users. The compromised data included personally identifiable, biological, and demographic information. The breach exploited a vulnerability that has not been disclosed by the software company. At least one journalistic source indicates that Illuminate "did not encrypt student information while the data was at rest" (The Record, 2022). Encrypting stored data, also known as data at rest, is known as a cybersecurity best practice, and was also mandated by their NYC Department of Education client (The Record, 2022).

The incident began on December 28, 2021, when a malicious actor gained unauthorized access to Illuminate's data. Illuminate detected and identified the unauthorized access, and by January 8, 2022, they had contained and/or eradicated it. During this time, they also removed access to the software for at least 10 days. They began alerting users of the incident starting March 25, and continued to disclose information until July 29, 2022.

Both the software producer and users suffered injuries and harms as net effects of the incident. The direct cost to Illuminate Education resulting from the breach is estimated at $22-30 million (NetDilligence Calculator, 2024; At-Bay.com Calculator, 2024). At the same time, the users lost their ability to use the software in some cases, and so had to use resources to find new solutions. Additionally, their personal data was compromised and can be used by nefarious actors, which could incur future harms and costs.

Illuminate, as a best practice and as required by their obligating regimes, would have had to notify users and regulators of the breach. This is a non-trivial process which would have required multiple people and offices within the organization, such as the cybersecurity team, legal counsel, public relations, and executive officers. The timeline for notifying users also varies by jurisdiction, and while notifications were sent out, some users complained that they were not notified in a timely manner (Class Action

Document 57 v. Illuminate Education, 2022). Following the formal notification of the breach, regulatory bodies would begin their investigation (The Record, 2022). Based on public information, this included the NYC DOE and other school districts, although few other regulatory bodies made their investigation public.

4 Responding + Recovering

In response to the incident, the stakeholders to the conflict sought to respond and recover (R&R) to the incident. They pursued strategies that might be achieved through various practices. While the end goals and specific tactics may have varied between the users, software producer, and regulators, they still followed a limited set of approaches. Specifically, there are four strategies that they pursued through various practices: posturing, compensating, correcting, and adhering/complying.

In pursuing the following strategies, each stakeholder had to make specialized decisions based on the variegated regimes that governed the conflict, and their individual needs in this context. There were intricate legal efforts that cannot be interpreted in this research. Additionally, there were choices that involved the input of multiple people from human resources, public relations, legal, IT, teams, that would have been formed totally internally and with many intricacies. The strategies would also have to take into account their relationships with the other stakeholders. Instead of trying to document the dynamics of these strategies in detail, I'm describing these practices abstractly as constructing a multistakeholder response. Each party, to achieve their best interests, would work with their networks to examine the regimes that have influence and assemble a legal vehicle that offered them the best opportunity to meet their goals, in the jurisdiction(s) where they would be most likely to succeed. They would use the below strategies together to construct a multistakeholder response.

4.1 Posturing

All stakeholders postured to protect their interests. The most prominent posturing was making public statements. Users, including end-users like students and teachers, school administrators, and school districts, as well as user representatives like advocacy groups and elected officials issued statements through internal channels and through the media (The Record, 2022; NYT, 2022). The software producer, Illuminate Education, made limited public statements (Greig, ZDNet, 2022; Elsen-Rooney, NY Daily News, 2022), presumably to protect their interests by limiting exposure, but must have spoken directly with their clients. Even their lack of public statements can be seen as posturing, as the company wanted to keep details hidden. Regulatory agencies, including local and national governments also issued statements about the attack. While the FTC did not respond directly to the Illuminate Education breach, they did issue an official statement and a Consumer Alert about education technology shortly after the breach was made public (FTC Policy Statement, 2022; FTC Consumer Advice, 2022; The 74, 2022). However, the NYC Mayor made more direct statements condemning the incident and the company's alleged failures (The Record, 2022).

Further posturing occurred when the parties formed and managed networks with other external stakeholders that had an interest in the conflict. This could include forming new networks, reinforcing existing ones, or even distancing themselves from stakeholders that posed a threat. On all sides lawyers and legal experts were consulted. The parties may have had internal legal counsel, but they also sought outside support. Parents of students began retaining lawyers to pursue legal action and Illuminate

Education worked with a separate law firm to represent them in court (Kisil v. Illuminate Education, 2022; Class Action Document 57 v. Illuminate Education, 2022; Class Action Document 75 v. Illuminate Education, 2023). Industry experts were also part of this network, providing valuable knowledge to the different parties, for example the K12 Security Information Exchange, a nonprofit focused on supporting cybersecurity in education technology.

The regulatory bodies would have also sought knowledge and expertise from outside parties to inform areas where they may not have enough in-house resources and gauge public interest. This might include collecting comments or amicus briefs and would have helped support their other strategies.

4.2 Compensating

All parties pursued compensation that was in their best interests. Users tried to maximize their compensation, while the software producer sought to reduce their costs to compensate users. Illuminate may have also sought monetary compensation if they had cybersecurity insurance. Regulators contributed to decisions on compensation for the affected parties, including the users and software producers. Regulators, including local and national government branches, pursued compensation according to their mandated responsibilities and jurisdiction on behalf of their constituents.

An example of the constructed response for compensation, multiple class action lawsuits were filed by parents of the student users in multiple courts (Kisil v. Illuminate Education; Class Action v. Illuminate Education). In the courts, because the case involved so many different jurisdictions, the parties had to evaluate their options, consolidate their arguments, and prove that they were owed compensation. They also worked with other parties and stakeholders to best achieve their goals. Eventually, they consolidated those lawsuits into a single filing, a process that required the input of the courts and the defendant, Illuminate Education. In the end, the court decided that the class did not have standing to sue: the response they constructed did not protect them such that they could receive compensation due to restrictions arising from the regime that governed their rights. Instead, the software producer's constructed response protected them from compensating the student users in court. In this situation, the regulatory stakeholder was the court, which ultimately made the decision.

The schools and districts that had formal agreements with Illuminate Education likely followed the terms of the agreements to seek redress. Compensation also would have included any terms that Illuminate Education and its clients agreed to when they signed their agreement. This could include the opportunity for the clients to opt-out of the agreement, and might also include reports or data that describe the incident so the clients can review it. No compensation for these harms has been disclosed publicly. It is possible that some clients may not have sought compensation.

4.3 Correcting

Each party had its own approach to the strategy of correcting the precursors and impacts of the incident. This means recovering from the event itself, as well as ensuring it doesn't happen again. Correcting is a separate strategy from compensating, in that it seeks to directly address the root causes and the negative impacts of the incident, so that operations can continue. Correcting therefore includes the immediate responses to remediate the issue and restore the compromised software, and business process changes that need to be made to prevent the incident from happening again. In this case, the software producer took their software offline for at least 10 days (Greig, 2022), but the broader business process changes are not known. It was alleged that Illuminate Education did not encrypt their data as

required in their agreements. For users, correcting might also include changes to business processes including reinforcing future agreements with other software producers and resolving problems that may have occurred due to the incident, such as a loss of access to the software. In this case, the user clients claim that Illuminate Education did not follow specific requirements regarding encryption, so another change to business processes might not be relevant for prevention.

For end-users, including teachers and students, and even school administrators, correction included a very different practice from those discussed thus far. To correct for the disruption caused by the breach, end-users who were required to desist from using the software (such as in NY), had to find new software solutions to meet their day-to-day needs. Far from seeking compensation in the courts, these users had to resolve software problems that were previously solved by Illuminate Education. Students, teachers, and administrators needed to continue to operate including managing grades and assignments without the PupilPath software. In spite of the breach, the users had interests that would have gone unaddressed without some software in place. These users deployed new software solutions, including creating new applications and procuring new solutions in order to meet their needs. This required the participation of higher-level constituents involved in the procurement processes including school districts and individual school administrators. In some cases, they would have implemented ad hoc solutions using readily available tools like those offered by Google or Microsoft. These ad hoc solutions might have compromised quality, meaning they weren't as capable at delivering the needs of the users, and even might have further compromised security if they weren't properly managed.

Regulatory stakeholders, including national and local government agencies and courts, also participated in correcting. These bodies seek correction by remediation of the impacts of the incident and helping to prevent such events from occurring in the future. However, they must take into account the interests of all the stakeholders. If they overcorrect, they may negatively impact the industry by dissuading future business. If they undercorrect, then they may not adequately protect the victims of the event. In either case, they may also set precedence for future action that has negative consequences. As such, the regulatory interventions must positively influence the software producer to comply with the rules without deterring them from conducting their business, while protecting the interests of the users so that they are protected from harms.

Initially, there were only two examples of regulatory correction in this case. The NYC DOE cut off all contracts, which led to other issues for the end users, but also set a precedent for the future. Also, the Student Privacy Pledge removed Illuminate from their program. However, in 2025, regulators announced punitive measures against Illuminate Education for the data breaches. Illuminate came to two separate settlement agreements, one with the FTC at the federal level, and another with a coalition of three state Attorneys General, New York, Connecticut, and California. The FTC settlement agreement included mandated security measures that would be enforced to ensure that Illuminate is in compliance (Merod, 2025; FTC, 2025). The settlement with the states included similar security measures as well as a $5.1 million penalty, to be distributed between the states (Troutman, Pepper, Locke, 2025). The joint enforcement between the states was likely the first action by a recently formed Coalition of Privacy Regulators, created to enforce state-level regulations for privacy violations (IAPP, LaCasse, 2025).

4.4 Adhering & Complying

Each stakeholder attempted to recover from the incident through adherence to and compliance with existing regimes.

The software producer was required to alert its customers in a set amount of time about the breach and some details about its effects. However, the rules about the notification and the details that needed to be included varied by jurisdiction, and failure to comply strictly with the rules is not necessarily going to warrant an enforcement action by the regulators. The breach consisted of a few days of unauthorized access by attackers to the software producers' data, which included the student data. In response, the timeliness, method, and information contained in the disclosure by the company would have been the subject of decisions based on a few factors: the various regime requirements under which it operated, their understanding of the incident, their capabilities or resources, and their evaluation of what would be best for the company itself. Ultimately, the disclosure would reflect their intentions toward adherence to the spirit of the regime, and their compliance with the written rules of the regime. The software producer disclosed the breach by notifying users starting March 25, 2022 through July 29, 2022 (NY Times, 2022; Kisil v. Illuminate Education, 2022; Class Action Document 57 v. Illuminate Education, 2022; Class Action Document 75 v. Illuminate Education, 2023). The network of users that filed the lawsuit argued that these notifications were not timely as required.

The users that were responsible for managing the relationship with the software producer also would also approach their response to the breach based on the rules of their respective regimes and the contracted agreements they had with Illuminate Education. In New York, they were able to terminate their business relationship. Other districts did not terminate business, and that could include limitations in their agreements and the laws surrounding their negotiations.

Regulatory response would have been managed by the offices with jurisdiction to respond to the issue. Their job is to ensure compliance with the law. However, they also have adherence and compliance strategies based on the interests of their appointed or elected officials, their resources, the capabilities of the team, and the strategic bureaucratic interests of the office, all of which factor into their constructed multistakeholder response. For example, the judge responsible for the lawsuit filed against Illuminate Education determined, based on law and precedent, that the case had to be dismissed.

5 Outcomes

The strategies and practices of the stakeholders to this conflict resulted in identifiable outcomes described in this section. These outcomes are comprised of outputs of their response and recovery practices, and also serve as inputs to the broader aftermath of the conflict.

5.1 Costs

For the users, one of the most visible costs resulting from the response to the breach was the need for new software solutions. In the New York jurisdiction, and possibly others, the severing of the relationship with Illuminate Education left gaps in the day-to-day operations of the users. This meant that users, with varying roles, had to seek out replacements to meet their needs. This ranged from implementing free software to developing new custom solutions, all of which have their own risks and costs associated with them. Free software may make their money by collecting the data of the users, leading to further privacy concerns. Creating a new solution requires capital and human resource investments. This includes time to develop the software and train users. In the meantime, the users are unable to adequately meet their needs, in this case issues like grading, viewing grades, managing student data, and more.

The most obvious cost to the users is that their data is now at risk of future harms. The personal data that was compromised can be used for cybercrime, which could lead to a variety of issues to those users affected.

All of the stakeholders incurred legal costs in this case. The users sued the company paid lawyers, the regulators used human resources to investigate and remedy it, and the company suffered reputational damage and lost contracts, as well as legal costs responding to the suits and investigations.

Based on the NetDiligence Data Breach Cost Calculator, the cost to the software producer is about $22 million. This includes costs associated with incident investigation, crisis management, regulatory fines & penalties, and lawsuits.

5.2 Transformations

The users of the Illuminate Education software implemented new software and adjusted their business practices. Higher level users likely addressed the requirements in their contracts to make more strict demands of their suppliers to prevent incidents like this from occurring in the future. All of these users are likely to have undergone further cybersecurity training to help with prevention as well.

The software producer, Illuminate Education, has been quiet about the event, so there is no data as to any transformation in their business processes. However, they were acquired by a private equity firm, Renaissance Learning, in August 2022, just 7 months after the breach (Kuykendall, 2022).

In New York, new rules and requirements were adopted by the primary regulatory body, Department of Education. The self-regulatory Student Privacy Pledge also transformed their enforcement approaches by removing a participant for the first time. Typically, they seek to encourage compliance with their rules (FPF, 2020), so removal is not an ideal response, it is better to punish signatories, but keep them within the fold. In this case, however, Illuminate Education was removed from the agreement.

In 2025, a group of state regulators joined together in a novel effort to pursue privacy regulation, and the first action they announced was a $5.1 million settlement with Illuminate Education (Troutman, Pepper, Locke, 2025). While this transformation took multiple years, the impact is broad, as it demonstrates the intense interest of state regulators in enforcing privacy laws. Crucially, the creation of the Consortium of Privacy Regulators occurred in 2025, years after the Illuminate data breach. At this point, that network formation might not even have registered as posturing for this particular case. However, by settling with Illuminate as their first (or one of their first) actions, they made a significant transformation to the enforcement approaches for privacy violations due to data breaches.

Courts at the national and state level did not make any transformations, instead they held to the precedent that individuals did not have a right of action to pursue compensation. Federal regulators in the FTC did come to a settlement agreement with Illuminate Education in 2025, but this represents standard operating procedure for consumer protection, not a transformation.

5.3 Preservation

The most apparent example of preservation in this case comes from the federal courts, which dismissed the class action lawsuit against Illuminate Education. The class action was the result of multiple lawsuits which were consolidated into a single complaint. However, the federal court decision to dismiss the complaint, based on their adherence with the existing laws, upheld the status quo making no changes to the regime.

The Student Privacy Pledge also sought to preserve their status as a self-regulatory body by removing the company. While in the past they have sought to encourage adherence, in this case they made a novel decision to remove the violating party (The 74, 2022).

There has also not been any public notice of action taken by some schools or school districts, indicating that they may have preserved their contracts with Illuminate Education in spite of the breach. This means that the users would also not have to make any changes to their operations.

5.4 Remedies

For the regulators, the efforts took time, but eventually resulted in penalties for the company. In the education jurisdictions, as we have seen, some regulatory bodies like the NYDOE canceled contracts with Illuminate Education, enforcing their contract rights. The Student Privacy Pledge removed the company from their agreement as an enforcement action. Later, in 2025, some of those states also punished Illuminate, resulting in them paying $5.1 million to New York, Connecticut, and California for violations of their privacy laws. The same year, the FTC also settled with the company, mandating cybersecurity upgrades.

Users and their advocates sought remediation through monetary compensation via the courts. More specifically, the parents of students filed lawsuits against Illuminate Education. They did not get any monetary compensation through this effort.

The courts, mainly the federal court, decided not to offer any remedy to the parties that sued.

6 Aftermath

Using the disruption, revision, change model of virtual weapon consequences (Kello, 2017), there do not seem to be any consequences of this case that impact the social cyber context. Regulators only executed existing, established procedures, resulting in no changes to the regimes, behaviors, or context that stakeholders observe.